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High-energy pulse self-compression and ultraviolet generation through soliton dynamics in hollow capillary fibres

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Optical soliton dynamics can cause extreme alteration of the temporal and spectral shape of a propagating light pulse. This occurs at up to kilowatt peak powers in glass-core optical fibres and at the gigawatt level in gas-filled microstructured hollow-core fibres. Here, we demonstrate optical soliton dynamics in large-core hollow capillary fibres. This enables scaling of soliton effects by several orders of magnitude to the multi-millijoule energy and terawatt peak power level. We experimentally demonstrate two key soliton effects. First, we observe self-compression to sub-cycle pulses and infer the creation of sub-femtosecond field waveforms—a route to high-power optical attosecond pulse generation. Second, we efficiently generate continuously tunable high-energy (1–16 μJ) pulses in the vacuum and deep ultraviolet (110 nm to 400 nm) through resonant dispersive-wave emission. These results promise to be the foundation of a new generation of table-top light sources for ultrafast strong-field physics and advanced spectroscopy.

Hollow capillary fibres (HCFs), consisting of a simple circular bore in a glass fibre, have been used in nonlinear optics since the 1970s[1,2]. As well as many other applications they have been used for frequency conversion and amplification through four-wave mixing[3,4], vacuum ultraviolet (VUV) generation[5], spatiotemporal self-compression[6–8], high-harmonic generation[9,10] and nonlinear optics in liquids[11]. In ultrafast optics their primary application is to pulse compression based on nonlinear spectral broadening[12,13]. Notably, there is one very important class of nonlinear effect that has not been demonstrated in gas-filled HCFs: optical solitons and the rich dynamics associated with them.

Bright temporal optical solitons are obtained by balancing a positive nonlinear refractive index with negative (anomalous) linear dispersion[14,15]. They underlie a wide range of important phenomena in nonlinear optics, especially in fibres. Some key examples are extreme pulse self-compression[16–20], the formation of white-light supercontinua[21] and the efficient generation of frequency-tunable pulses through resonant dispersive-wave (RDW) emission[17,18,22–27]. These effects have been thought impossible to achieve in HCFs in the visible and near-infrared (NIR)[28,29] due to their weak dispersion in these spectral regions.

The group velocity dispersion (GVD) of the HE_{nm} modes of a gas-filled hollow fibre is given by

$$\beta_2(\lambda) \approx \frac{\lambda^3}{4\pi c^2}\left(\rho_r \frac{\partial^2 \chi_e}{\partial \lambda^2} - \frac{u_{nm}^2}{2\pi^2 a^2}\right) \quad (1)$$

where n and m are the mode indices, u_{nm} is the mth zero of the Bessel function J_{n-1}, c is the speed of light, a is the HCF core radius, λ is the wavelength, ρ_r is the gas density relative to some standard conditions and $\chi_e(\lambda)$ is the linear electric susceptibility of the gas under those standard conditions (see Supplementary Information). As the first term in equation (1) is positive at optical wavelengths for nearly all gases, negative GVD can only be obtained using the second term, which describes the dispersion of the waveguide. Importantly, this waveguide contribution is weaker for larger core sizes.

Hollow-core photonic-crystal fibres (HC-PCFs) can offer low-loss guidance despite their small core radius (typically around 15 μm), leading to sufficient anomalous waveguide dispersion for soliton effects to be observed at low pulse energy[28–32]. In comparison, the loss of a conventional HCF with such a small core is prohibitively high[33]. The $1/e$ power loss length is given by

$$L_{loss} \approx \frac{4\pi^2 a^3}{3u_{nm}^2 \lambda^2} \quad (2)$$

which is just 1 cm for the HE_{11} mode in a 15-μm-radius HCF at a wavelength of 800 nm.

The use of HCF with much larger core radii (~75–500 μm) reduces the loss dramatically ($L_{loss} \propto a^3$) and also allows for much higher pulse energy and peak power than in HC-PCF. However, it strongly reduces the waveguide dispersion, and so the gas nonlinearity dominates. In conventional HCF pulse compression, propagation in the hollow fibre is used only to increase the pulse bandwidth through nonlinear self-phase modulation (SPM) and self-steepening, and dispersion in the HCF is neglected. Instead, linear dispersive mirrors after the fibre provide the negative quadratic phase required to temporally compress the pulse (we refer to this as post-compression). This is shown schematically in Fig. 1a. In this way compression to few- and even single-cycle pulses can be achieved at high pulse energies up to a few millijoules[13,34–39].

Theoretical suggestions of how to obtain soliton effects in HCFs—which require nonlinearity and dispersion to act simultaneously within the fibre—have included the use of metal-coated HCFs to reduce the loss[40], pumping in higher-order modes to increase u_{nm} in equation (1)[41,42], or moving to longer wavelengths[43,44]. However, only numerical results on self-compression have been reported so far, with the rich variety of soliton-induced effects remaining unexplored, and no experiments performed to date.

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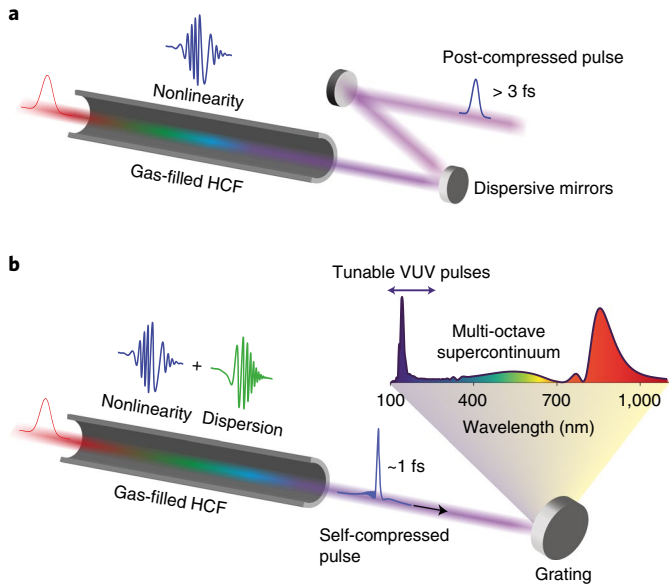

Fig. 1 | Comparison between post-compression and soliton dynamics in HCF. a, Conventional post-compression in a gas-filled HCF, where the dynamics in the HCF are dominated by nonlinear spectral broadening based on self-phase modulation and self-steepening, and the anomalous dispersion for pulse compression is provided by chirped mirrors. **b**, Soliton dynamics in a gas-filled HCF, where the dynamics in the HCF combine both nonlinearity and dispersion, leading directly to extreme self-compression, ultraviolet emission and multi-octave supercontinuum formation.

In this Article we determine the general scaling rules for soliton dynamics in HCFs and find that they can in fact be broadly accessed in the fundamental mode and without any coatings or microstructure, even in the visible and NIR. This is possible simply by using shorter pump pulses, longer HCF lengths than usual—enabled by the important innovation of HCF stretching[36,37,39,45,46]—or both of these approaches simultaneously. We experimentally demonstrate two key effects, as illustrated in Fig. 1b. We observe soliton self-compression to sub-cycle pulses with 43 GW peak power, and soliton-driven ultraviolet pulse generation between 110 nm and >350 nm with energies of 1–16 μJ. These results represent a power and energy scaling of up to 1,000 times compared to previous results in HC-PCF. Our numerical simulations—extensively verified by experiment—predict that these dynamics can be further scaled to achieve self-compression at the terawatt level, providing a route to optical attosecond pulses[47] with unprecedented power, as well as table-top generation of tunable and ultrafast near-ultraviolet to VUV (3–12 eV) pulses at energies over 100 μJ, with a peak brightness exceeding free-electron laser systems.

Results

Scaling rules for soliton dynamics in HCF. The character of ultrashort pulse propagation depends on the full wavelength-dependent dispersion landscape and the balance between nonlinearity and dispersion. This balance is characterized by the pump soliton order $N=(L_d/L_{nl})^{1/2}$, where the dispersion length L_d and the nonlinear length L_{nl} describe the characteristic length scales of GVD and SPM, respectively[48]. In an HCF a reasonable parametrization of the dispersion landscape is the relative location of the pump wavelength λ_0 with respect to the zero dispersion wavelength λ_{zd} (defined via $\beta_2(\lambda_{zd})=0$). This determines the phase-matched location for RDW generation and therefore, in the usual case in which λ_0 is fixed, changing λ_{zd} (via the gas pressure) tunes the RDW emission wavelength. It also determines the degree of high-quality self-compression that can be achieved. For the strongest unperturbed compression we require a broad anomalous dispersion region, with λ_{zd} tuned far from λ_0.

Figure 2 shows numerically modelled propagation of 10 fs pulses at 800 nm in helium-filled hollow fibres with very different core sizes (see Methods). For each core size we tune the gas pressure so that $\lambda_{zd}=380$ nm (matching the self-compression experiments described later), and adjust the energy to maintain the soliton order at $N=2.6$. Consequently we obtain nearly identical temporal and spectral dynamics even though the energy is increased by almost three orders of magnitude (0.006 mJ to 3 mJ). The small differences—apart from the length scale over which the dynamics occur—are due to different propagation losses in each case. This scaling is a manifestation of the general scaling of nonlinear optics in gases[49], which—when applied to hollow fibres—states that if one scales the core radius by a factor η, exactly the same dynamics (excluding losses) occur if one also scales the gas pressure by $1/\eta^2$ and the fibre length and peak power by η^2.

The low-energy results in Fig. 2a correspond to an idealized small-core anti-resonant guiding HC-PCF without any loss. In real HC-PCF, cladding resonances inherent to their guidance mechanism would create multiple bands of large loss and dispersion[50], especially in the VUV part of the spectrum. Nevertheless, dynamics similar to those in Fig. 2a have been previously demonstrated[17–20,23–25,27] and utilized in both angle-resolved photoemission spectroscopy[26] and time-resolved photoelectron imaging spectroscopy[51]. Figure 2b,c shows results for simple, non-microstructured, HCFs. The larger core sizes in HCF enable significant energy scaling.

Self-compression is observable in Fig. 2 (marked i) because the pump pulse parameters correspond to a high-order soliton ($N>1$) propagating in the negative dispersion region. Therefore, SPM initially dominates, broadening the pulse spectrum; this enhances the effect of the negative GVD, which compensates for the nonlinear phase and results in temporal compression of the pulse[14,16,21,48] and subsequent soliton fission dynamics[21,52–54] (see Supplementary Fig. 1 for a spectrogram depiction of these dynamics). In the line plots in Fig. 2 we see that this self-compression produces 1 fs pulses. For the small-core case this 1 fs pulse reaches a peak power of 2 GW, whereas in the largest core it is scaled to 1.2 TW.

In the spectral domain, the self-compressed pulse corresponds to a supercontinuum spanning four octaves, starting from 100 nm (blue spectral line plots). The subsequent spectral dynamics show efficient RDW emission around 130 nm for all core sizes (marked ii), with VUV energies of 0.6, 25 and 300 μJ for Fig. 2a–c, respectively. We extract the VUV pulses by spectral filtering (Supplementary Fig. 2). They reach maximum peak powers of 0.2, 6 and 80 GW, with a pulse duration of 1.8 fs and a bandwidth limit of 0.8 fs.

Due to the high intensities reached in Fig. 2—up to 8×10^{14} W cm^{-2} in all cases—as much as 0.3% of the gas is ionized (leading to a 40% energy loss) and plasma effects play a role. In particular, a soliton self-frequency blueshift, along with corresponding soliton acceleration (identified by the curved trajectory of the soliton in the time-domain plots marked iii), is evident[55,56], and these both shift and enhance the strength of the VUV RDW emission[25,56]. These effects are not required for self-compression and RDW emission, both of which still occur with ionization and plasma effects turned off in the simulations (Supplementary Fig. 3), or for other sets of parameters that require lower pulse intensities (higher pressures and longer λ_{zd}).

Figure 2 is a specific example of soliton dynamics in HCF. In general we would like to vary the pump pulse duration, the zero dispersion wavelength λ_{zd} (which tunes the RDW emission wavelength[18,23]) and the core size (to control the energy and power level). To ensure that the soliton dynamics are not precluded by the HCF loss, we need to maintain $L_{fiss}<L_{loss}$, where $L_{fiss}\approx L_d/N$ is the soliton fission length, which approximates the required length scale for soliton self-compression and subsequent pulse

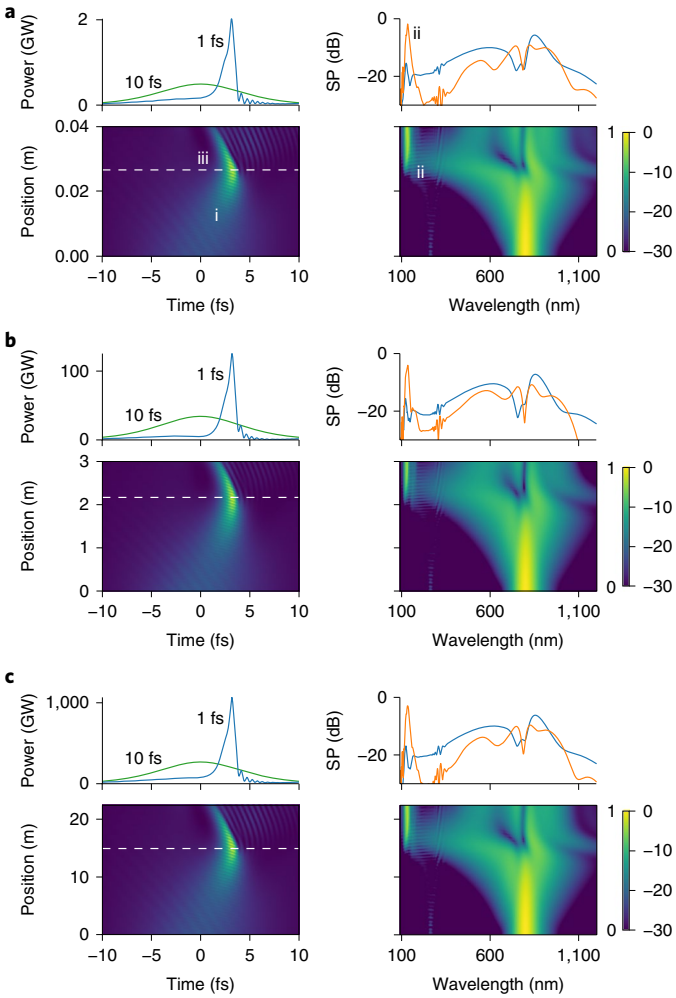

Fig. 2 | Scaling of soliton dynamics in gas-filled hollow fibres. Numerically modelled propagation of 10 fs, $N = 2.6$ solitons at $\lambda_0 = 800$ nm, in hollow fibres filled with helium such that $\lambda_{zd} = 380$ nm. The colour density plots show the evolution of the temporal (linear scale) and relative spectral power (SP) (dB scale). The time-domain line plots show the input pulse (green) and the maximally self-compressed pulse (blue), which occurs at the position marked with the white dashed lines. The blue spectral line plot shows the corresponding supercontinuum spectrum and the orange spectral line plot shows the spectrum at the end of the propagation. **a**, Example of dynamics in a small core, similar to HC-PCF (but neglecting resonances) with no loss: $a = 15$ µm, 27.8 bar helium, energy of 0.006 mJ. **b**, HCF with $a = 125$ µm, 400 mbar helium, energy of 0.4 mJ. **c**, HCF with $a = 350$ µm, 51 mbar helium, energy of 3 mJ. See Supplementary Fig. 1 for a spectrogram depiction of the dynamics.

breakup[21]. It can be shown (see Supplementary Information) that in HCF this length is

$$L_{\text{fiss}} \approx \frac{a^2 \tau_{\text{fw}}^2}{3N\ |\delta(\lambda_0, \lambda_{\text{zd}})|} \propto \frac{a^2 \tau_{\text{fw}}}{\sqrt{I_0}} \qquad (3)$$

where τ_{fw} is the full-width pulse duration at half maximum power (FWHM), I_0 is the peak intensity and $\delta(\lambda, \lambda_{\text{zd}})$ is a function that describes the dispersion of the filling gas.

Figure 3 shows this scaling rule for $\lambda_0 = 800$ nm as a function of core radius and pump pulse duration for two example values of λ_{zd}. The minimum L_{fiss} is obtained for each pulse duration by finding the

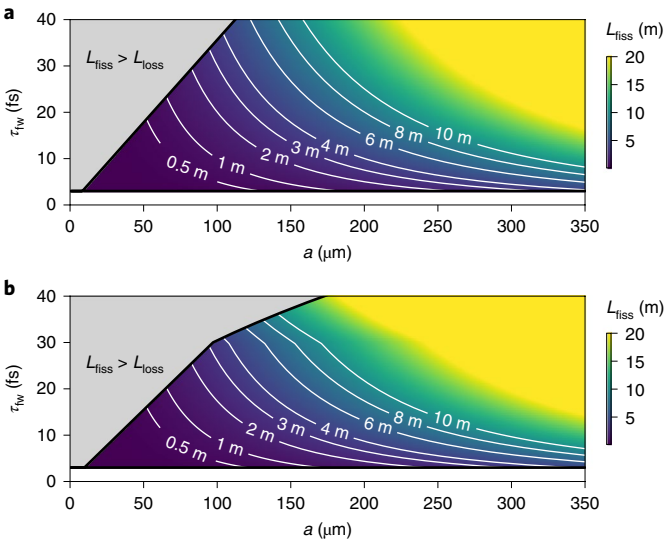

Fig. 3 | Soliton fission length scaling in helium-filled HCFs. a,**b**, The contours and colour scale show the minimum soliton fission length L_{fiss}, calculated from equation (3) using the largest permissible N, as a function of core radius a and pulse duration τ_{fw} for $\lambda_0 = 800$ nm and $\lambda_{\text{zd}} = 380$ nm (**a**) and $\lambda_{\text{zd}} = 700$ nm (**b**). In the grey shaded region it is impossible to obtain $L_{\text{fiss}} < L_{\text{loss}}$.

maximum N or I_0, which is determined by several factors. First, to obtain coherent soliton self-compression as opposed to incoherent modulational instability dynamics we must ensure $N < 15$ (ref. [21]). Second, N increases with the peak power and intensity of the pump pulse, but these are limited by self-focusing[57] and ionization, respectively (see Supplementary Information).

The grey regions in Fig. 3a,b show where it is impossible to observe soliton dynamics because the propagation loss is too high and $L_{\text{fiss}} > L_{\text{loss}}$. This occurs for small core sizes and long pulse durations. However, $L_{\text{fiss}} \propto a^2$, whereas $L_{\text{loss}} \propto a^3$ (equations (2) and (3)), so it is always possible to obtain $L_{\text{fiss}} < L_{\text{loss}}$ by moving to larger core sizes (right-hand side of Fig. 3a,b), at the cost of increasing the length of HCF required. This, in turn, can be remedied by pumping with shorter pulses—an initial duration of 10 fs or less keeps the fission length below 3 m for core radii up to $a = 200$ µm, and even below half a metre for the smallest core sizes.

Until recently, most conventional HCF post-compression systems were limited to HCF lengths of ~1 m. This is much shorter than the fission length for the 30 fs pump pulses at 800 nm that are typically used, which explains why soliton fission dynamics have not been observed in HCF previously.

High-energy soliton self-compression. As an initial experiment to demonstrate self-compression in HCF we used the pulses directly produced by our Ti:sapphire system (see Methods) which have 26 fs duration. We used a 3-m-long stretched HCF with 125 µm core radius. At a maximum energy of 400 µJ coupled into the HCF filled with 1 bar of helium ($\lambda_{\text{zd}} = 470$ nm), the soliton order is $N = 6.8$ and the corresponding fission length is 4.2 m. Although the HCF is too short to reach the fission point, Fig. 4a shows clear experimental evidence of self-compression from 26 fs to 9 fs as we increase the pump energy (and hence the soliton order). Furthermore, our numerical modelling reproduces the experimental data extremely closely, both in terms of the power spectrum (Fig. 4b) and the fine details of the pulse shape in the time domain (Fig. 4a). Such good agreement is rarely achieved, and it supports both the accuracy and validity of our numerical model and the fidelity of the pulse measurement.

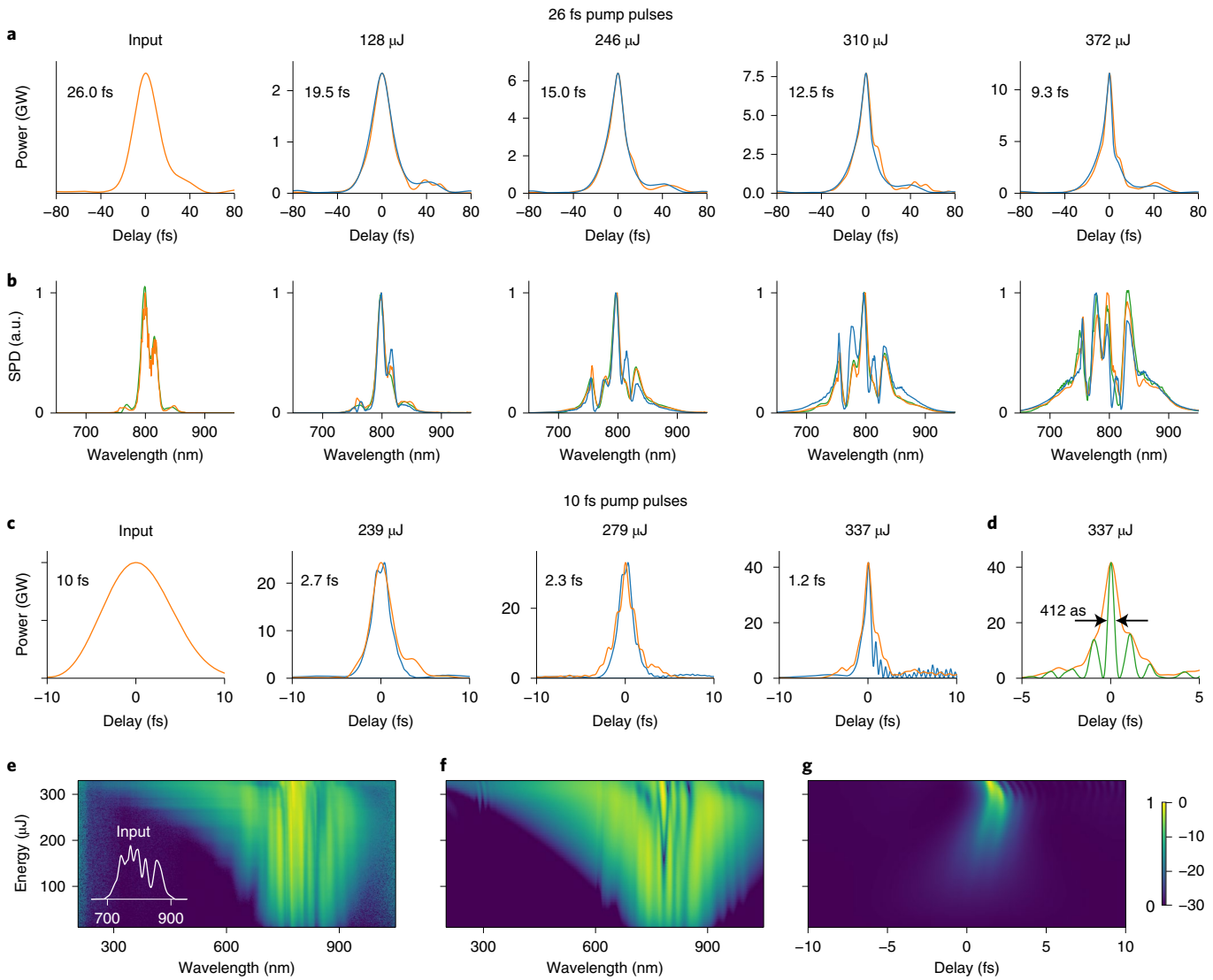

Fig. 4 | Soliton self-compression in gas-filled HCFs. a,**b**, For 26 fs pump pulses in 1 bar helium filling gas. **a**, Temporal pulse compression for increasing pump energy: orange, measured from second-harmonic frequency-resolved optical gating (SHG-FROG) pulse retrieval (see Supplementary Fig. 4 for details); blue, numerically modelled. **b**, Corresponding pulse spectra: orange, FROG retrieved; blue, numerically modelled; green, independently measured. **c**–**g**, For 10 fs pump pulses in 0.4 bar helium filling gas. **c**, Temporal pulse compression for increasing pump energy: orange, measured from PG-FROG pulse retrieval (see Supplementary Fig. 5 for details); blue, numerically modelled. **d**, Closer view of **c** but with the square of the retrieved electric field shown in green. Note the limitations of the measurements underlying the pulse shapes shown in **c** and **d** as described in the main text. **e**, Measured optical spectrum for increasing pump energy. Inset: the pump spectrum on a linear scale. **f**, Corresponding numerically modelled spectrum. **g**, Corresponding numerically modelled temporal pulse shape. Spectra in **e** and **f** are on a 30 dB log colour scale, and in **g** on a linear colour scale. All results are for a 3-m-long HCF with 125 μm core radius.

To observe complete soliton self-compression we start with 10 fs pump pulses produced in a conventional HCF post-compression set-up (see Methods). Figure 4c–e shows the experimental results. At a helium pressure of 0.4 bar (λ_{zd} = 380 nm) and an energy of 239 μJ (N = 2) we obtain self-compression from 10 fs to 2.7 fs, which is close to a single cycle and slightly broader than the bandwidth-limited duration of 2.5 fs. The pulse becomes even shorter as we increase the energy, compressing to 2.3 fs (bandwidth limit 2.1 fs) at 279 μJ and finally, at 337 μJ, to 1.2 fs for a bandwidth limit of 1.1 fs. At this point, the full width of the square-field profile, shown in Fig. 4d, is only 412 as—the soliton has self-compressed to an optical attosecond pulse[47].

These pulse measurements are based on a new polarization-gating frequency-resolved optical gating (PG-FROG) technique that covers the band from 200 nm to >1,100 nm (see Methods), supporting the 1 fs sub-cycle pulse bandwidth. Nevertheless, it must be noted that the technique has some severe limitations and the sub-cycle pulse duration cannot be reliably established from our PG-FROG measurements alone. The pulse profiles shown in Fig. 4 are obtained by measuring the pulses after they have been stretched by propagation through dispersive materials—most notably the exit window of the HCF system and air in the laboratory—and subsequently numerically back-propagating them to the exit of the HCF. In general this process is rigorous, with the refractive index of air, silica and MgF_2 well-characterized. For the 2.7 fs pulse the back-propagated pulse duration changes by less than 5% within the experimental uncertainty on the propagation distance and window thickness. However, the extreme bandwidth and extension into the

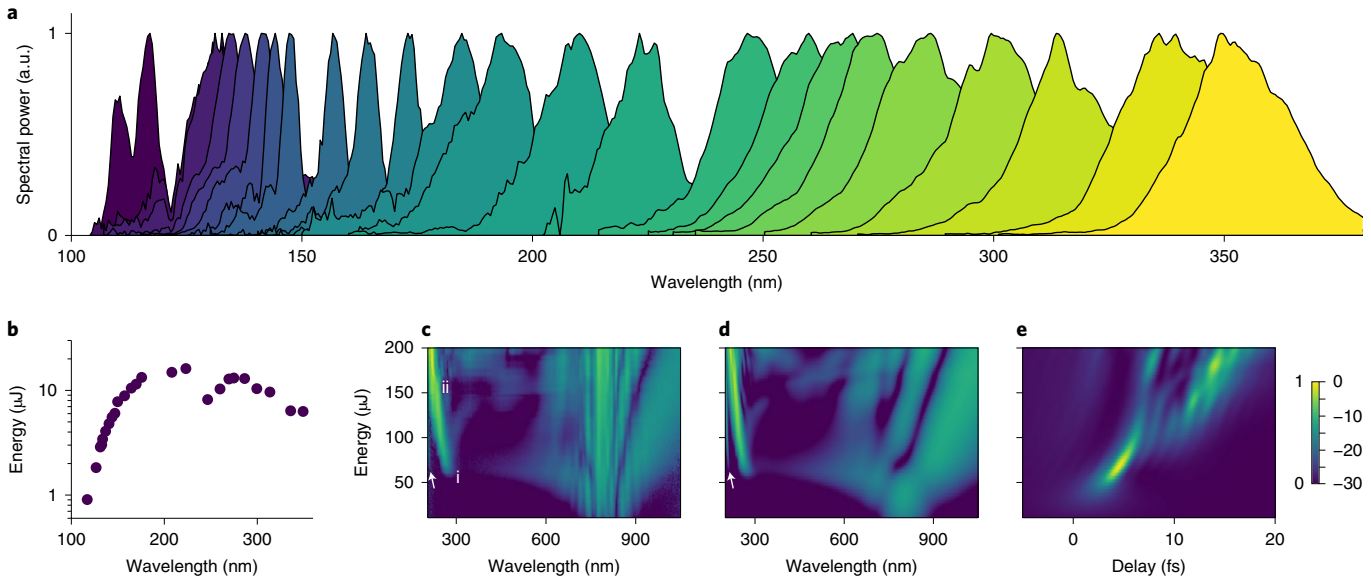

Fig. 5 | Tunable RDW emission in gas-filled HCFs. a, Measured RDW spectra from a 3-m-long HCF with 125 μm core radius filled with helium and pumped with 10 fs pulses (the pump spectrum is shown in the inset to Fig. 4e). The pressures used range from 230 mbar (shortest wavelength spectrum) to 4 bar (longest wavelength spectrum). Full parameters are listed in Supplementary Table 1. Extension to beyond 400 nm was also measured at higher pressures. **b**, Measured VUV to DUV pulse energies. **c**, Spectral evolution at the output of the HCF for 10 fs pump pulses as a function of input pulse energy for 1 bar neon. **d**, Corresponding numerical simulation. **e**, Corresponding numerical simulation of the temporal power. Plots in **c** and **d** are on a 30 dB log colour scale; **e** is on a linear colour scale. White arrows in **c** and **d** indicate the RDW emission in the HE_{12} mode.

ultraviolet of the sub-cycle pulses means that a difference of only 10 μm in glass thickness is sufficient to stretch the 1.2 fs pulse by a factor of 2.5. Because PG-FROG always requires a transmissive interaction medium, even placing the apparatus into vacuum would not avoid this issue. Fully independent experimental evidence will only be provided by much more complicated experiments such as attosecond streaking[47,58].

There are two additional observations that clearly point towards the generation of optical attosecond pulses in our experiments. The first is the consistent and excellent agreement between all of our experimental data and the numerical model. This is demonstrated in Fig. 4a,b and again for the particular case of sub-cycle self-compression in Fig. 4e,f, where we numerically reproduce the spectral power density, allowing us to infer the temporal dynamics (Fig. 4g). The simulated and measured pulse profiles shown in Fig. 4c are also in excellent agreement despite the uncertainties of the measurement. The second observation is that compression to a sub-cycle pulse is required for RDW emission to occur in the VUV wavelength range. This can be understood from the cascaded four-wave-mixing picture of RDW emission[59] in which all of the frequencies between the RDW and the pump need to be present simultaneously. That this is the case is clearly apparent from the spectrograms in Supplementary Fig. 1. For RDW emission in the VUV we have the simultaneous presence of frequencies spanning up to three octaves, which corresponds to a sub-cycle pulse in the time domain. The correlation between VUV RDW emission and sub-cycle pulse generation has also been found in HC-PCF[24,25]. Therefore, our measurements of RDW emission across the VUV (see below) also support the existence of such extreme pulse compression in HCF; in particular, the parameters for the 1.2 fs pulse in Fig. 4c are the same as for the 130 nm RDW shown in Fig. 5a.

Tunable vacuum and deep ultraviolet pulse generation. At the extreme compression point, the soliton breaks up due to higher-order linear and nonlinear effects, such as self-steepening, shock formation and higher-order dispersion[21,52–54]. The latter induces RDW emission—often referred to as emission of fibre-optic Cherenkov radiation or resonant radiation[17,18,22,23,25,53,54,59,60]. Simulations of the process in HCF are shown in Fig. 2, with RDW emission occurring at the point marked ii. Further details of the RDW emission dynamics are provided in the spectrograms in Supplementary Fig. 1. The emission wavelength depends on the dispersion landscape and hence on the gas pressure and core size (equation (1)).

Figure 5a shows the experimental tuning of RDW emission from below 110 nm to 350 nm by changing the gas pressure in our 125 μm HCF pumped with 10 fs pulses (the full parameters are provided in Supplementary Table 1). Further tuning beyond 400 nm was also obtained at higher pressures. Due to the larger core size, the emitted pulses in the VUV are up to 1,000 times more energetic than obtained in HC-PCF, as shown Fig. 5b and Supplementary Tables 1 and 2. Even at the shortest RDW band, which spans 107–121 nm, the energy is approximately 1 μJ; this increases to around 13 μJ near 180 nm, 16 μJ near 220 nm and remains around 10 μJ until starting to decrease at 300 nm due to the lower pump energy required at higher pressures. The estimated conversion efficiency to the VUV ranges from 0.5% around 120 nm up to 5% around 170 nm and in the deep ultraviolet (DUV, 200 nm to 300 nm) this increases to over 15% (Supplementary Table 1). Upscaling of the system as a whole would provide much higher pulse energies at all ultraviolet wavelengths (Fig. 2).

The RDW emission process is highly stable—we measure a relative intensity noise of 1.4% at the 252 nm RDW peak (Supplementary Fig. 6). Furthermore, we have not observed any degradation of transmission through the HCF over 15 months of prolonged DUV and VUV generation experiments because the high-intensity ultraviolet light is confined in the hollow core and does not significantly overlap with the glass cladding, whereas the exit window (when used) can be damaged after prolonged exposure to high DUV fluence.

Similar results can be obtained in other gases with the pressure adjusted for their different dispersion and nonlinearity. As one example, Fig. 5c shows the experimental RDW emission dynamics

at 1 bar of neon (λ_{zd} = 550 nm). The RDW wavelength changes from 275 nm (marked i) to 225 nm (marked ii) as the pump energy increases from 60 to 200 μJ (from N=2.3 to N=4.3). This is a well-known effect arising from the nonlinear contribution to the phase-matching[23,61]. Figure 5d shows the corresponding numerical simulations, which once again closely reproduce the experiment, including fine details at the −30 dB level over the full range of the energy scan. Figure 5e shows the corresponding simulations in the time domain, illustrating the initial self-compression process followed by soliton fission.

The RDWs in Fig. 5a,c are in the fundamental mode of the HCF. We confirmed this by measuring the divergence of the filtered RDW emission at 355 nm and comparing it to the numerically calculated divergence of the fundamental HCF mode. The results are shown in Supplementary Fig. 7, and exhibit almost exact agreement. Weaker RDWs in the HE_{12} mode are also generated and appear as additional features in the spectra (for example, as shown by the white arrows in Fig. 5c,d) at the corresponding phase-matched point[62].

Discussion

Advances in the control and measurement of the dynamics of matter with light have been contingent upon our ability to produce light pulses with decreasing temporal duration and wider spectral coverage. We have shown that soliton dynamics in gas-filled HCF significantly expands our toolset in both directions.

While post-compression can be optimized to produce high-quality single-cycle pulses[38], shorter pulses require the use of multiple chirped-mirror sets[47,58]. Soliton self-compression offers a much less complex and lower-cost route to sub-cycle pulse generation. It can be used in any spectral region, whereas chirped mirrors are designed and manufactured for a specific wavelength range. By replacing the conventional pre-compressor in our set-up with an additional soliton self-compression stage, as discussed in ref. [63], the requirement for chirped mirrors could be removed entirely. Two-stage HCF systems, as used here, have previously been demonstrated for enhanced non-soliton spectral broadening in HCF[64,65], and are readily implemented in laboratories with existing HCF post-compressors. Furthermore, soliton self-compression can be energy-scaled with the fibre core size, the only cost being the increased laboratory length required. As described in this Article, this offers a route to terrawatt-level sub-femtosecond pulse generation, and to optical attosecond pulses 20 times more powerful than state-of-the-art light-field synthesizers[47]. The fission length in such a system is around 15 m, as shown in Fig. 2c. While long, this would be a unique source of 1 fs, 1 TW pulses, and the apparatus would still be smaller than other facility-scale lasers. As described in equation (3), using 5 fs pulses (available from current systems) would halve the fission length to 7.5 m. That this length scale is practically feasible has already been demonstrated in conventional compressors using 6 m HCF[39].

The ability to scale soliton dynamics by two or more orders of magnitude in HCF goes far beyond pulse compression. As we have demonstrated, we can use soliton-driven RDW emission as a table-top source of tunable DUV/VUV pulses with high energy and few-cycle duration. The DUV energies demonstrated here are ten times higher than the best results from HC-PCF[27], and the VUV energies demonstrated here are up to three orders of magnitude higher[25]. In both cases the conversion efficiency was higher. Furthermore, HCF is free of the guidance resonances that are inherent to HC-PCF and which compromise the guidance of UV radiation and reduce the lifetime of the waveguide. We also achieve two orders of magnitude higher conversion efficiency as compared to previously demonstrated schemes of high-energy VUV generation[5,66–68]. Those schemes were all pumped with much higher energies, achieved similar or lower output energy, could not be tuned as widely (if at all) and produced VUV pulses that were longer than we expect to have produced here (see Supplementary Table 2 for a detailed comparison). Recent experiments in HC-PCF have shown that RDW emission in the DUV can be extremely short in duration, generating pulses as short as 3 fs (refs. [69,70]). Given the close agreement between our numerical simulations and experiments, we can provide insight into the temporal structure of the VUV RDW (which is beyond the spectral range of our pulse measurement techniques) and confirm the short pulse duration for VUV generation in gas-filled HCF. For example, the VUV pulse generated in the simulations in Fig. 2b has a duration of just 1.8 fs when it has the highest peak power of 6 GW (Supplementary Fig. 2). For diffraction limited beams the peak brightness is proportional to peak power (see Supplementary Information) and so our RDW emission source has a similar peak brightness to VUV generation in free-electron lasers, which produce up to gigawatt VUV peak power[71,72]. In the predictions for the larger-core HCF (Fig. 2c), the VUV pulse energy is increased to over 300 μJ and the peak power to over 80 GW, thus providing a table-top source with ten times the peak brightness of a free-electron laser in the VUV.

That the full range of soliton dynamics can be accessed in HCF is in itself a remarkable discovery. Beside sub-cycle self-compression and RDW emission, other phenomena already observed in HC-PCF could also be transferred to HCF, such as Raman-soliton effects[24,73], soliton-plasma dynamics[55], among many others[28,29,31,32].

The new light source of high-energy sub-cycle pulses and tunable high-brightness VUV–DUV pulses presented here, along with the rich dynamics from additional soliton effects, yet to be explored in HCF, should form the basis for a new generation of experiments in ultrafast science, advanced time-resolved spectroscopy, and nonlinear optics pumped in the VUV.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-019-0416-4.

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Author contributions
J.C.T. proposed this work, the initial theory and the experimental design. He also performed the numerical simulations and drafted the manuscript. All authors contributed to the experimental implementation and refinement, the analysis and discussion of the results, and the editing of the manuscript.

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The authors declare no competing interests.

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Methods

Main experimental set-up. The pump pulses were produced by a commercial Ti:sapphire oscillator, regenerative amplifier and single-pass amplifier chain (Coherent Legend Elite Duo USX) producing 26 fs, 800 nm pulses at 1 kHz repetition rate. The experimental set-up is depicted in Supplementary Fig. 8. In a first stage, the 26 fs pump pulses were compressed using an HCF post-compression set-up, consisting of a home-built 1.6 m stretched capillary system[45,46], with 225 μm inner radius, filled with helium (with a typical throughput at full power, including coupling and transmission, of >70%), followed by 12 reflections from double-angle chirped mirrors (PC70, Ultrafast Innovations), a broadband attenuator based on a $\lambda/2$ waveplate and Brewster-angle reflection from a silicon plate, and anti-reflection coated BK7 wedges for dispersion fine-tuning (Femtolasers). This system was tuned to compress the pulses at the entrance of the second-stage HCF. It was set to either deliver 26 fs (when the first stage HCF was evacuated) or around 10 fs bandwidth-limited pulses when the first-stage helium pressure was 2.2 bar.

The subsequent soliton stage consisted of a home-built 3 m stretched HCF system with 125 μm inner radius. The coupling efficiency to this stage is 60–70% and the linear transmission is 65% at 800 nm. The gas pressure could be controlled, and was stable to, within 0.1 mbar. The windowless output of this stage was directly connected to a vacuum system, containing a fully calibrated, home-built, VUV spectrometer, consisting of either a 1,200G/mm Al + MgF_2 coated, or 2,400 G/mm Pt-coated, concave aberration corrected diffraction grating (McPherson), two 200 μm slits and a VUV silicon diode (SXUV100G, OptoDiode), which has been calibrated at the Physikalisch-Technische Bundesanstalt, Germany (PTB). Alternatively, the gratings in our VUV spectrometer can be shifted under vacuum to allow the output of the HCF to pass through our vacuum system, either through a 2 mm silica or 1 mm MgF_2 output window, to be measured in atmosphere with an integrating sphere coupled to a DUV-to-NIR charge-coupled device (CCD) based spectrometer (calibrated as a system), a camera for beam profiling, a calibrated thermal power meter or two different FROG set-ups.

VUV pulse energy measurements. The absolute spectral energy measurements are achieved by collecting the entirety of the HCF output beam with the VUV grating and focusing the diffracted beam through a large aperture onto our calibrated diode. The aperture was set to ensure only the VUV part of the spectrum was collected. The spectral energy density (in μm nm^{-1}) was obtained from the photocurrent as follows: the photocurrent was filtered to obtain the zero frequency (d.c.) component and processed to account for the pump repetition rate, the absolute diode calibration, the gain of our current amplifier (DLPCA-200, FEMTO Messtechnik) and the manufacturer-supplied theoretical grating efficiency (which provides a conservative lower bound to the estimated energy density, as the actual grating efficiency is always lower than the theoretical one). Measurements with a silica or BK7 window as a filter were also used to verify the VUV signal below 160 nm and DUV signal below 280 nm, and exclude re-entrant spectra or scatter from the zeroth-order diffraction.

The absolute calibration was verified by independently measuring a RDW generated at 240 nm with 1.8 bar helium with both the VUV spectrometer and our DUV-to-NIR CCD spectrometer with integrating sphere; the latter was calibrated for absolute spectral irradiance with a PTB traceable deuterium lamp and quartz-tungsten halogen lamp traceable to the United States National Institute of Standards and Technology (NIST). Furthermore, we measured the diffracted DUV signal directly with an additional calibrated photodiode (Ophir PD300-UV). The results all agreed with each other to within 6%, which is within the stated uncertainty of the calibration sources themselves. The measured energy was 8.5 μJ, in agreement with Fig. 5b.

Pulse characterization measurements. *Second-harmonic FROG.* Pulse characterization of pulses longer than 9 fs was based on a home-built all-reflective split-mirror second-harmonic FROG device, based on a 10-μm-thick beta barium borate (BBO) crystal cut for type I phase-matching. The raw traces were spectrally corrected using the frequency marginal and retrieved using the extended ptychographic iterative engine (ePIE)[74]. The retrievals were checked to be independent of the different initial conditions and recovered the independently measured spectrum (Supplementary Fig. 4). The resulting pulses were then back-propagated to account for the air path and 2 mm glass window at the output of the second capillary stage. The excellent agreement between simulation and pulse measurements (Fig. 4) verifies that this process was reliable.

PG-FROG. Pulse characterization of pulses shorter than 9 fs was carried out using a novel ultrabroadband PG-FROG device, the layout of which is shown in Supplementary Fig. 9. The raw traces were corrected for the spectral response of the apparatus as well as the frequency-dependent nonlinearity by using the frequency marginal and the separately measured power spectrum of the unknown pulse. When taking into account the known spectral sensitivity of the spectrometer and the theoretical frequency dependence of the efficiency of the nonlinear process, the frequency marginal of the trace already closely matches the measured spectrum; however, the correction aids the pulse retrieval process. The raw traces contained an additional signal at the wavelength of the gate pulse caused by scatter in the fused-silica sample. This was removed by high-pass filtering the raw traces in this wavelength region, attenuating delay-independent features. The pulse retrieval was also carried out using ePIE[74], with the modification that, at each iteration, the retrieved gate pulse was updated by numerically back-propagating the retrieved test pulse to the gas cell exit and then forward-propagating it through the optics in the gate pulse arm of the device. Initial guesses for the test and gate pulses were formed by numerically propagating a transform-limited pulse from the capillary exit to the interaction point, taking into account the air path as well as the optical elements. This significantly improved both the speed at which the retrieval converges and the retrieval error reached at convergence. Supplementary Fig. 5 shows the measured and retrieved traces. The resulting pulses were then back-propagated to account for the air path and optics to the output of the second capillary stage.

Numerical simulations. We performed rigorous numerical propagation simulations using our unidirectional, full-field, spatiotemporal propagation code. Our model includes a full vector polarization model, intermodal coupling, modal dispersion and loss, the Kerr effect and self-focusing, as well as photoionization and plasma dynamics.

Our numerical simulations agree extremely well with experimental data without the use of any adjustable parameters; all material properties are based on the best available values[25,75,76], the stretched HCF is almost exactly modelled by the Marcatili model we employ[33] and the input pulse parameters are based purely on our experimental measurements.

We expand the full vectorial spatiotemporal electric field in the frequency domain and over the natural hollow-fibre modes

$$\mathbf{E}(t,r,\theta,z)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\mathrm{d}\omega\sum_j\hat{\mathbf{e}}_j(r,\theta)E_j(\omega,z)\exp(-i\omega t) \tag{4}$$

where r and θ indicate the transverse position in the hollow fibre core, z the axial position in the fibre, t represents time, ω is the angular frequency, j is the mode index, $\hat{\mathbf{e}}_j$ are the orthonormal transverse electric field shapes of the modes[33] and E_j are the complex spectral amplitudes of the modes, which evolve during propagation. Following refs. [62,77], the spectral amplitudes are coupled via

$$\partial_z E_j(\omega,z)=\left(i\beta_j(\omega)-i\frac{\omega}{v}-\frac{\alpha_j(\omega)}{2}\right)E_j(\omega,z)+i\frac{\omega}{4}P_j^{\mathrm{nl}}(\omega,z) \tag{5}$$

$$P_j^{\mathrm{nl}}(\omega,z)=\int_0^{2\pi}\mathrm{d}\theta\int_0^a r\,\mathrm{d}r\;\hat{\mathbf{e}}_j^*(r,\theta)\cdot\mathbf{P}^{\mathrm{nl}}(\omega,r,\theta,z) \tag{6}$$

where β_j and α_j are the mode propagation and attenuation constants[33] (see Supplementary Information), v is a suitably chosen, but arbitrary, reference frame velocity, and $\mathbf{P}^{\mathrm{nl}}(\omega,r,\theta,z)$ is the Fourier transform of the full vector and spatially resolved nonlinear polarization.

Our model proceeds by numerically integrating equation (5). At each step we make use of equation (4) to obtain the full electric field, from which we calculate the full nonlinear polarization for projection back onto the modes through equation (6). This projection is accurately computed using the full two-dimensional integral over the cross-section of the fibre core using a parallel p-adaptive cubature method.

The nonlinear polarization term, for the purposes of the present Article, includes the Kerr effect through

$$\mathbf{P}^{\mathrm{Kerr}}(t,r,\theta,z)=\rho_{\mathrm{r}}\varepsilon_0\chi^{(3)}[\mathbf{E}(t,r,\theta,z)\cdot\mathbf{E}(t,r,\theta,z)]\,\mathbf{E}(t,r,\theta,z) \tag{7}$$

where ε_0 is the electric permittivity of free space, $\chi^{(3)}$ is the third-order nonlinear susceptibility of the filling gas at some standard conditions[76] and ρ_{r} is the gas density relative to those conditions. The ionization and plasma response is included through[78]

$$\begin{aligned}\mathbf{P}^{\mathrm{ion}}&(t,r,\theta,z)\\&=I_{\mathrm{p}}\int_{-\infty}^{t}\mathrm{d}t'\frac{\partial_t n_{\mathrm{e}}(r,\theta,z)}{|\mathbf{E}(t',r,\theta,z)|}+\frac{e^2}{m_{\mathrm{e}}^2}\int_{-\infty}^{t}\mathrm{d}t'\int_{-\infty}^{t'}\mathrm{d}t''n_{\mathrm{e}}(r,\theta,z)\mathbf{E}(t'',r,\theta,z)\end{aligned} \tag{8}$$

where the free electron density is given by

$$n_{\mathrm{e}}(r,\theta,z)=n_0\left(1-\exp\left[-\int_{-\infty}^{t}\mathrm{d}t'\,w(|\mathbf{E}(t',r,\theta,z)|)\right]\right) \tag{9}$$

I_{p} is the ionization potential of the filling gas, e is the electron charge, m_{e} the electron mass, n_0 the initial neutral gas density and $w(|\mathbf{E}(t,r,\theta,z)|)$ is the Perelomov, Popov, Terent'ev (PPT) ionization rate[79,80].

Our code runs in parallel; for this Article each simulation was run over 72 CPU cores. The radial integrals are adaptively evaluated to ensure convergence at each step.

The pump pulse initial conditions were always for a linearly polarized pulse in the fundamental HE_{11} mode. At the output the polarization extinction ratio

was always larger than 300 dB (our numerical accuracy), indicating that no depolarization occurs in ideal conditions.

We include the modes HE_{1m} with m typically up to 10, and both polarization orientations for each mode. We have verified, through extensive convergence tests, that no other mode types and symmetries are excited, and that no energy is transferred to any higher-order HE_{1m} modes. In fact minimal energy is transferred to modes higher than HE_{11}. What is transferred primarily consists of energy at specific high-order-mode phase-matched RDWs (Fig. 5c,d)[62].

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Code availability

The computer code used in this study will be made available upon reasonable request to the corresponding author.

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